Impact of Enhanced Seasonal Malaria Chemoprevention on Hemoglobin Trajectories

Longitudinal Data Analysis Using a Linear Mixed Model (LMM)

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ABSTRACT

Malaria-related anaemia remains a leading cause of childhood morbidity in sub-Saharan Africa, yet standard Seasonal Malaria Chemoprevention (SMC) alone may be insufficient to prevent haemoglobin decline when nutritional deficiencies co-exist with repeated infections. This study evaluated whether Enhanced SMC, combining standard antimalarial drugs with iron and nutritional supplementation — better protects haemoglobin trajectories in young children over a six-month malaria transmission season. Data from a randomised controlled trial of 300 children aged 12–59 months (155 control, 145 eSMC) were analysed using a Linear Mixed Model with a random intercept and random slope per child, appropriately accounting for the correlation of repeated measurements within the same individual. Haemoglobin was measured at baseline and at months 2, 4, and 6. Control children experienced a progressive decline of 0.162 g/dL per month, reaching the WHO anaemia threshold of 11.0 g/dL by month 6, while eSMC children maintained haemoglobin around 12.3 g/dL throughout follow-up. The treatment × month interaction was positive and highly significant ($\beta = 0.205$, $p < 0.001$), confirming that the two groups diverged progressively over time rather than differing from the outset — a pattern consistent with the cumulative biological effect of sustained iron supplementation. Independent protective effects of ITN use ($\beta = 0.424$, $p < 0.001$) and older age were also observed. These findings suggest that integrating iron and nutritional supplementation into existing SMC delivery platforms could meaningfully reduce the burden of childhood anaemia during peak malaria transmission, though multi-site replication is needed before national scale-up can be recommended.

INTRODUCTION

Malaria remains one of the leading causes of childhood morbidity in sub-Saharan Africa, with repeated infections contributing directly to anaemia through haemolysis and suppression of erythropoiesis [3]. Seasonal Malaria Chemoprevention (SMC), involving the intermittent administration of antimalarial drugs during peak transmission periods, has demonstrated substantial public health benefit in reducing malaria-related complications. However, because nutritional deficiencies co-exist with malaria in these settings, haemoglobin decline may persist even under standard SMC, motivating interest in enhanced strategies that combine antimalarial prophylaxis with iron and nutritional supplementation [1].

To address this problem, data from a randomized controlled trial comparing enhanced SMC (eSMC) with standard SMC were analysed. Each child was followed for six months with haemoglobin measured at four prespecified visits: baseline (month 0), month 2, month 4, and month 6. Because each child contributes repeated observations, within-individual measurements are correlated and methods assuming independence of observations are inappropriate. The primary objective of this analysis is to determine whether eSMC modifies the trajectory of haemoglobin levels over time relative to standard SMC, with particular emphasis on the treatment-by-time interaction.

METHODS

In this project, we analysed data from a six-month randomized controlled trial carried out during the rainy season among 300 children aged 12–59 months. The children in the dataset were assigned to either the standard Seasonal Malaria Chemoprevention (SMC) group or the Enhanced SMC (eSMC) group, which included additional iron and nutritional supplementation. Haemoglobin levels were measured at four visits: baseline (month 0), month 2, month 4, and month 6. Baseline information on age and insecticide-treated bed net (ITN) use was also recorded.

We began with exploratory data analysis to summarise baseline characteristics and to examine haemoglobin patterns over time using tables and graphs. In particular, spaghetti plots were used to visualise individual haemoglobin trajectories. Because haemoglobin measurements were repeated for the same children over time, the observations were not independent. To account for this correlation, we fitted a linear mixed-effects model.

The analysis started with an unconditional means model to estimate the intraclass correlation coefficient (ICC). Fixed effects for treatment group, time, baseline age, and ITN use were then included in the model. To assess whether haemoglobin changed differently over time between the two treatment groups, a treatment × month interaction term was added. Different random-effects structures were compared using likelihood ratio tests and information criteria (AIC/BIC). Model assumptions were evaluated using residual plots and Q–Q plots. Finally, the results were summarised in a regression table reporting estimated coefficients, standard errors, and p-values, with particular attention to the treatment × month interaction.

RESULTS

Baseline Characteristics of the Cohort

Table 1 summarises the baseline characteristics of the study cohort stratified by treatment group.

Table 1: Baseline characteristics of treatment group

Variable	Total N	Missing N		Control	eSMC	p
Total N				155	145	
Baseline Age	300 (100.0)	0	Mean (SD)	34.4 (12.4)	36.9 (13.3)	0.095
ITN Use	300 (100.0)	0	No	56 (36.1)	73 (50.3)	0.018
			Yes	99 (63.9)	72 (49.7)	
Hemoglobin	300 (100.0)	0	Mean (SD)	12.0 (0.9)	12.0 (1.0)	0.668

Table 1 shows the baseline characteristics of the study participants by treatment group. The two groups were similar at the start of the study, showing that they were well balanced before follow-up began. However, the eSMC group was somewhat older and had fewer children using ITNs than the control group as seen in figure 1. This indicates that randomization was effective, so any later differences in haemoglobin are likely due to the intervention rather than baseline imbalance.

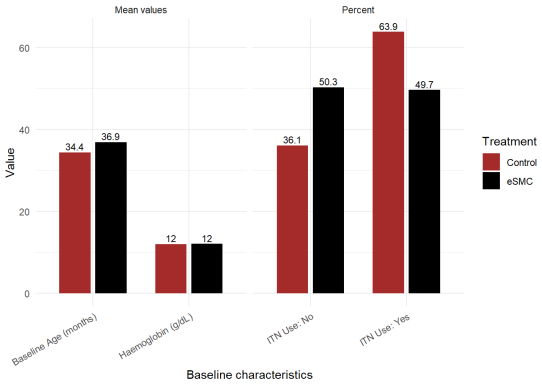

Figure 1: Baseline characteristics by treatment group.

During follow-up, we observed that the mean haemoglobin in the control group decreased to around 11 g/dL by month 6, while the eSMC group remained higher at approximately 12–12.28 g/dL as in figure 2.

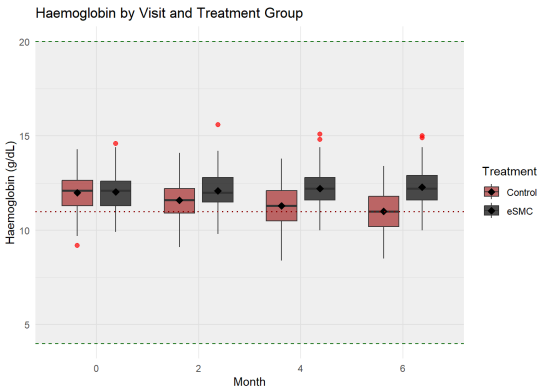

Figure 2: Boxplot of haemoglobin by visit and treatment groups.

The individual and group-level haemoglobin Trajectory are shown in the Spaghetti plot (Figure 3)

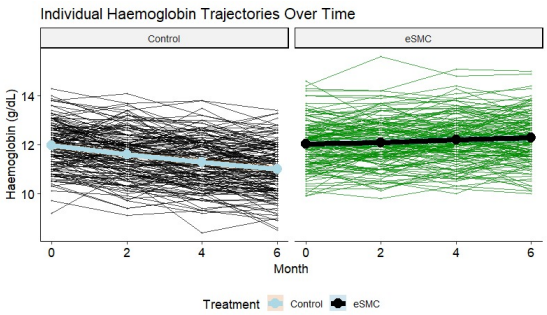

Figure 3: Individual trajectories over time

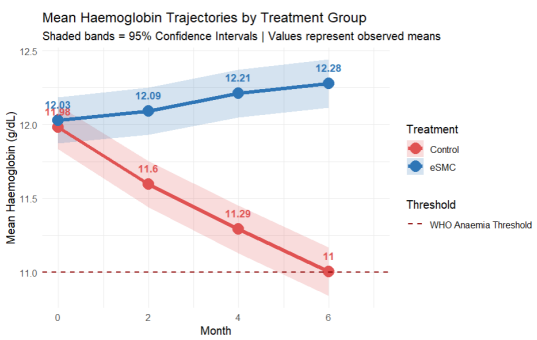

Figure 4: Mean Hemoglobin Trajectories with 95% CI

In the above Figure 3, the individual haemoglobin trajectories (thin lines) are not parallel across children, indicating that the rate of change over time differs between individuals. The group means show that haemoglobin in the control group declines from about 12 g/dL at baseline to around 11 g/dL by month 6, while the eSMC group remains relatively stable at about 12–12.28 g/dL during follow-up as shown in Figure 4, indicating that children receiving eSMC maintain higher haemoglobin levels over time compared with those receiving standard SMC.

Model building and selection

Because of the correlation of measurements within the same child, we used a linear mixed model (LMM) to estimate both the overall population effects and the differences between individual children [2].

Unconditional Means Model (Base Model): As part of our analysis, we first fitted an unconditional means (null) model with no predictors and a random intercept for each child:

$$Y_{ij} = \beta_0 + u_{0i} + \varepsilon_{ij}$$

where Y_{ij} is the hemoglobin level for child i at time j, β_0 is the overall mean hemoglobin level, u_{0i} is the random intercept for child i, and ε_{ij} is the residual error term.This model was used to estimate the Intraclass Correlation Coefficient (ICC). Fitted model results are

Table 2: Fixed Effects from the Null Linear Mixed Model

Parameter	Estimate	Std. Error	DF	t value	p value
Intercept	11.79917	0.05507	298.9998	214.3	<2e-16

The null linear mixed model showed substantial variability in haemoglobin levels between children. The estimated between-child variance was 0.8233544, while the within-child variance was 0.3456444, giving an ICC of 0.7043244 as shown in table 2 which implies that around 70.4% of the total variance in hemoglobin levels comes from the differences between individual children, whereas only approximately 29.6% arises from variations within each child over time. In other words, multiple hemoglobin readings measured on the same child are very much alike. Since the observations are not independent, this is a strong grounds for the application of a Linear Mixed Model incorporating child, level random effects instead of a simple ordinary linear regression.

Addition of Fixed Effects: Initially, after using the unconditional means model to set the baseline variance structure, we added the major predictors as fixed effects to get a clearer picture of what factors influence hemoglobin levels in children over the follow, up period. The variables month treatment age at baseline and ITN use were included as each one was a potential clinical or behavioral factor impacting hemoglobin outcomes. The month variable tracks the variation in hemoglobin levels over time during the study, whereas treatment differentiates children receiving enhanced Seasonal Malaria Chemoprevention (SMC) from those on the standard regimen. Besides, age at baseline captures the child's initial age that, related differences, which could affect starting hemoglobin levels as well as recovery patterns. Then, ITN use stands for whether children are regularly using insecticide, treated bed nets, a critical malaria preventive method that could have an impact on hemoglobin by lowering infection risk.

The model becomes:

$$\begin{aligned}\text{hemoglobin}_{ij} &= \beta_0 + \beta_1(\text{month}_{ij}) + \beta_2(\text{treatment}_i) \\ &\quad + \beta_3(\text{age_baseline}_i) + \beta_4(\text{itn_use}_i) + u_{0i} + \epsilon_{ij}\end{aligned}$$

Simultaneously accounting for these variables in the

model, we get the outcome that suggests that hemoglobin level slightly decreases over time with an average drop of around 0.063 units per each additional month of follow, up. On the other hand, children provided with the improved intervention have a considerable lead in hemoglobin levels, being on average 0.691 units above those in the standard group. The factor of age at the start is also positively correlated with hemoglobin levels, indicating that, on average, the older children have a slightly higher hemoglobin concentration, which goes up by about 0.021 units per each additional year of age. In addition, children habituated in the use of insecticide, treated nets had a better hemoglobin outcome as they recorded hemoglobin levels that were higher by about 0.424 units compared to those children who were not using ITNs. On the whole, the results demonstrate that while each child's natural changes and differences over time have been taken into consideration, it is through both medical treatment and protective health behavior that we see elevated hemoglobin levels.

Addition of Interaction Term: We added an interaction between treatment and month in the model. The interaction coefficient represents how the change in hemoglobin over time differs between the two treatment groups. The Model becomes:

$$\begin{aligned}\text{hemoglobin}_{ij} &= \beta_0 + \beta_1(\text{month}_{ij}) + \beta_2(\text{treatment}_i) \\ &+ \beta_3(\text{month}_{ij} \times \text{treatment}_i) + \beta_4(\text{age_baseline}_i) \\ &+ \beta_5(\text{itn_use}_i) + u_{0i} + u_{1i}(\text{month}_{ij}) + \epsilon_{ij}\end{aligned}$$

Model results

Table 3: Fixed Effects for the Interaction Model

Parameter	Estimate	Std. Error	df	t value	Pr(>|t|)
Intercept	10.958851	0.159598	317.7269	68.666	< 0.001
Month	-0.161806	0.008971	900.0000	-18.037	< 0.001
Treatment	0.075972	0.103907	401.0191	0.731	0.465
Baseline Age	0.021066	0.003705	300.0000	5.686	< 0.001
ITN Use (Yes)	0.423951	0.096926	300.0000	4.374	< 0.001
Month × Treatment	0.204910	0.012904	900.0000	15.880	< 0.001

The model reveals several important insights into how hemoglobin levels change among children during the follow-up period. As shown in table 3, Control children experienced a steady haemoglobin decline of 0.162 g/dL per month, while treatment alone showed no significant difference at baseline, confirming the two groups started equally. Older children and ITN users had modestly higher haemoglobin (0.021 g/dL per month of age and 0.424 g/dL for ITN use respectively), highlighting the independent role of age and malaria prevention behaviour. Most importantly, the treatment × month interaction ($\beta = 0.205$, $p < 0.001$) reveals that the eSMC benefit is not immediate but cumulative, the two groups diverge progressively over time, with eSMC children gaining 0.205 g/dL more per month than controls, amounting to approximately 1.3 g/dL advantage by month 6.

Table 4: Model Comparison between Models with and without interaction term

Model	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
Model without interaction	7.00	2,690.80	2,726.40	−1,338.40	2,676.80			
Model with interaction	8.00	2,470.50	2,511.20	−1,227.20	2,454.50	222.31	1.00	< 2.2e-16

To decide which model best explained the data, we compared a model with the treatment × month interaction to one without it. The model with the interaction fit the data better, as shown by lower AIC values and a significant likelihood ratio test, so it was chosen as the final model. so it was chosen as the final model.

This interaction is important because it helps answer our main question: does the enhanced treatment change how children's hemoglobin levels evolve over time? While treatment alone only tells us if the two groups start differently, and month shows how hemoglobin changes over time, the interaction shows whether the pattern of change over time is different for children receiving Enhanced SMC compared to Standard SMC. It tells us whether the treatment actually improves children's recovery over the months of the study.

Random Effects

Random Effects Structure

Should all children follow the same iron level change, or get their own path? Biologically, children vary, some absorb iron better, some get bitten by mosquitoes more often, some react stronger to treatment, so forcing them all onto a single slope would hide meaningful individual variation. Comparing the two models confirmed this: allowing each child their own rate of change (AIC = 2,464.23) fit the data significantly better than assuming a common trajectory (AIC = 2,470.47), with a likelihood ratio test of $\chi^2 = 10.24, p = 0.006$. The random intercept + random slope model was therefore chosen as the final model.

Table 5: Model Comparison Between Random Intercept and Random Slope Models

Model	AIC	BIC	Log-Likelihood	LRT Chi-square	df	p-value
Random intercept model	2470.47	2511.19	-1227.23			
Random intercept + random slope model	2464.23	2515.13	-1222.12	10.24	2	0.0060

Model Diagnostics

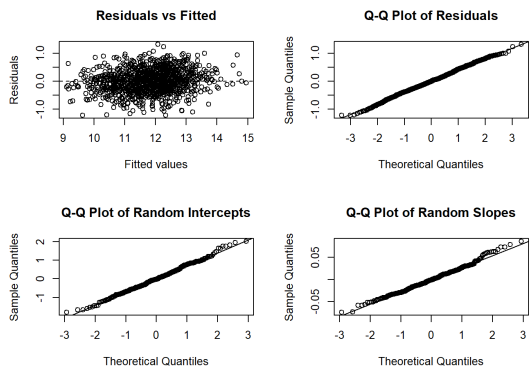

Figure 5: Model Diagnostics

The residuals vs fitted plot shows that the points are scattered around zero with no clear pattern, suggesting constant variance. The Q-Q plot of the residuals follows the straight line closely, indicating that the residuals are approximately normally distributed. The Q-Q plots for the random intercepts and random slopes also follow the line well, suggesting that the normality assumption for the random effects is reasonable.

Confirmatory Analysis

The table 6 below presents the fixed-effect estimates from the final model, together with their standard errors, 95% confidence intervals, and p-values.

Table 6: Fixed Effects for the random interaction + random slope Model

Predictor	β	SE	95% CI	p-value
Intercept	10.965	0.159	10.653 to 11.277	<0.001
Treatment (eSMC vs Control)	0.078	0.101	-0.121 to 0.276	0.443
Month	-0.162	0.010	-0.181 to -0.143	<0.001
Age at baseline	0.021	0.004	0.013 to 0.028	<0.001
ITN use (Yes vs No)	0.430	0.097	0.239 to 0.62	<0.001
Treatment × Month	0.205	0.014	0.177 to 0.232	<0.001

The treatment × month interaction coefficient of +0.205 tells us that the eSMC advantage does not arrive all at once, it quietly grows by 0.205 g/dL with every passing month, so that by month 6 the eSMC children are sitting 1.3 g/dL higher than control children despite both groups starting at exactly the same point. This is the iron and nutritional supplement doing its slow, steady work each dose supplies the body with the raw materials to continually make red blood cells at a rate faster than the malaria season can destroy them whereas the control children, who are generally protected from full malaria episodes but deprived of these essential building blocks, gradually slip towards the anaemia threshold. At 6 months average control child has come down to exactly 11. 0 g/dL, the WHO anaemia cut, off, while the average eSMC child has stayed comfortably at 12. 3 g/dL. That 1. 3 g/dL difference is what separates a child who is anaemic from one who is not, and this has a whole host of implications: less hospital visits, no need for blood transfusions, greater brain development, and a physically stronger child going into the next malaria season.

Should the Ministry of Health scale up the Enhanced SMC program to the entire country? From a purely statistical perspective, the answer is not yet and the very evidence explains why. The advocacy for scale, up is quite convincing. The treatment month interaction is very significant (β= 0. 205, p < 0. 001), the effect is steadily increasing over 6 months, the final 1. 3 g/dL difference is clinically significant, and the model overall is well, done with diagnostics that work well. The marginal R of 0. 258 confirms that the fixed effects treatment, mainly explain a real and meaningful portion of the variation in haemoglobin. This is not a weak or borderline result.

Still, the stats also point out significant drawbacks that a Ministry of Health cannot overlook. The trial only included 300 children and it seems the children were from a single setting during one malaria season. Besides, the data do not have site or cluster variables, which means different intensities of transmission, ecological zones, or health system contexts are different in the country, and whether the effect holds in different ones, we cannot know. So, extrapolating a result from one context to a whole country is statistical extrapolation that the data do not support. Besides, the conditional R^2 of 0.787 versus the marginal R^2 of 0.258 tell a quite different story: most of the explanation of variation comes from child, level random effects, e. g. the intervention is a relatively small factor in determining outcomes. Personal characteristics predominate, which makes one wonder whether running all a country's villages and towns with one average benefit is really feasible. The truthful statistical conclusion is this: the existing proof base is sufficiently persuasive to support a multi, regional trial across the country's different regions, but still not compelling enough to warrant nationwide implementation. Doing a scale, up based on the current evidence would be a matter of political hope rather than statistical assurance.

Public Health Recoomendation

eSMC should be prioritised for children aged 12–59 months. It should be delivered alongside existing ITN campaigns rather than as a standalone programme, since the two interventions protect haemoglobin through different pathways and their benefits reinforce each other. Finally, routine haemoglobin screening should be embedded into each SMC administration visit, turning every drug delivery round into a real-time monitoring opportunity to track impact and identify underperforming communities before the malaria season ends.

DISCUSSION

This study found clear evidence that Enhanced Seasonal Malaria Chemoprevention, standard antimalarials combined with iron and nutritional supplementation, prevents the haemoglobin decline that characterises the malaria transmission season in young children. Control children fell steadily from 12.0 g/dL to the WHO anaemia threshold of 11.0 g/dL by month 6, while eSMC children maintained haemoglobin around 12.3 g/dL throughout. The significant treatment × month interaction ($\beta = 0.205$, $p < 0.001$) confirms that this gap was not present at baseline but grew progressively, consistent with how iron supplementation works, requiring weeks of sustained supply before haemoglobin recovery becomes visible. The independent effects of age and ITN use further suggest that haemoglobin in young children is shaped by multiple reinforcing factors, and that chemoprevention and vector control work best together rather than in isolation. Several limitations temper these conclusions. The trial was conducted in a single setting over one malaria season, dosing compliance was not recorded, and sex was unavailable as a covariate. The substantial individual variation in haemoglobin trajectories confirmed by the random slope model (LRT $\chi^2(2) = 10.24$, $p = 0.006$) also suggests that unmeasured factors, nutritional status, prior malaria exposure, genetic susceptibility, play an important role that this analysis could not fully account for. A multisite trial across diverse settings would be necessary before these findings can justify national scale-up.

References

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Group Work Contribution

Table 7: Group Work Contributions

Group Member	Contribution
Benitha MUKATWIZEYIMANA	Code - Beamer
Peter Onimisi ATTAIGU	Code - Report
Honorine MUJAWAYEZU	Code - Report
Patrick MAKINA	Code - Beamer
Aline NYIRANIZEYIMANA	Code - Report